

Ischemic Stroke Lesion Segmentation

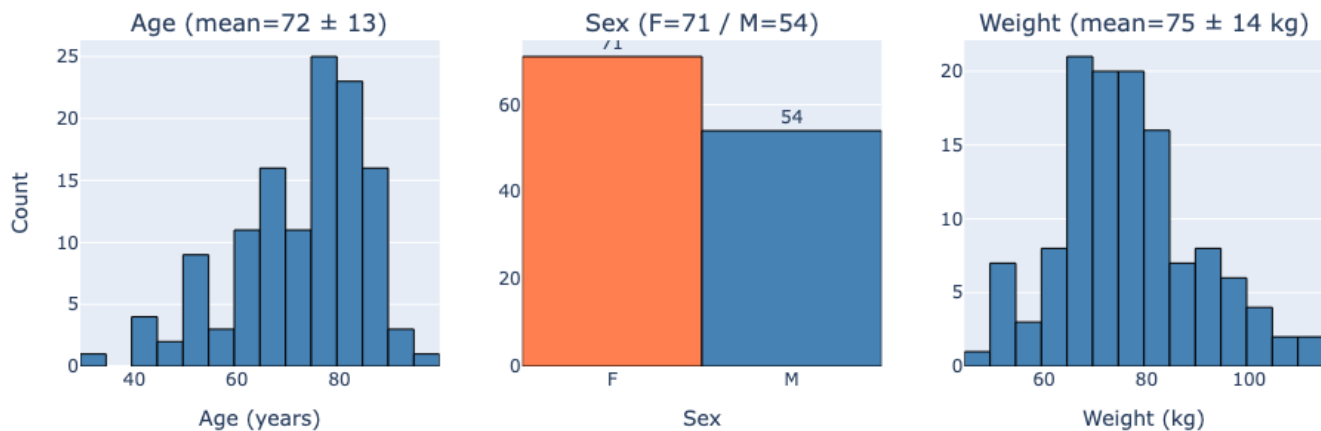
Kajenthini Kobivasan
CO4 Project - FS26



Dataset & Task

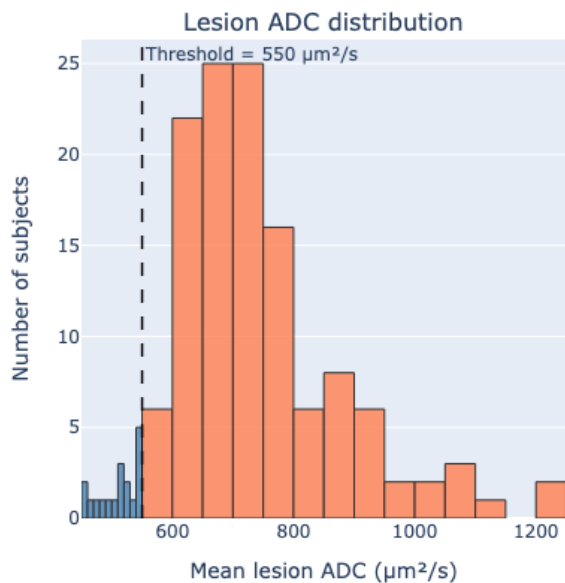
- ISLES-2022: 250 cases, ADC maps + expert masks
- Goal: segment ischemic lesion automatically

ISLES-2022 demographics (n=125)

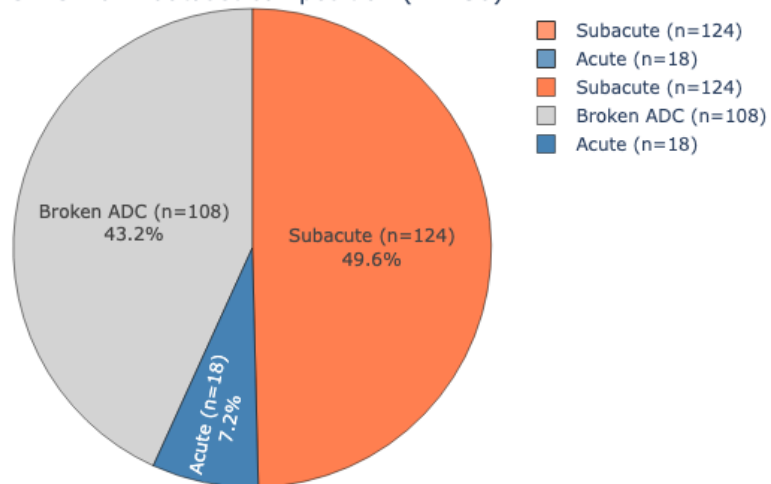


Dataset & Task

ISLES-2022 stroke phases

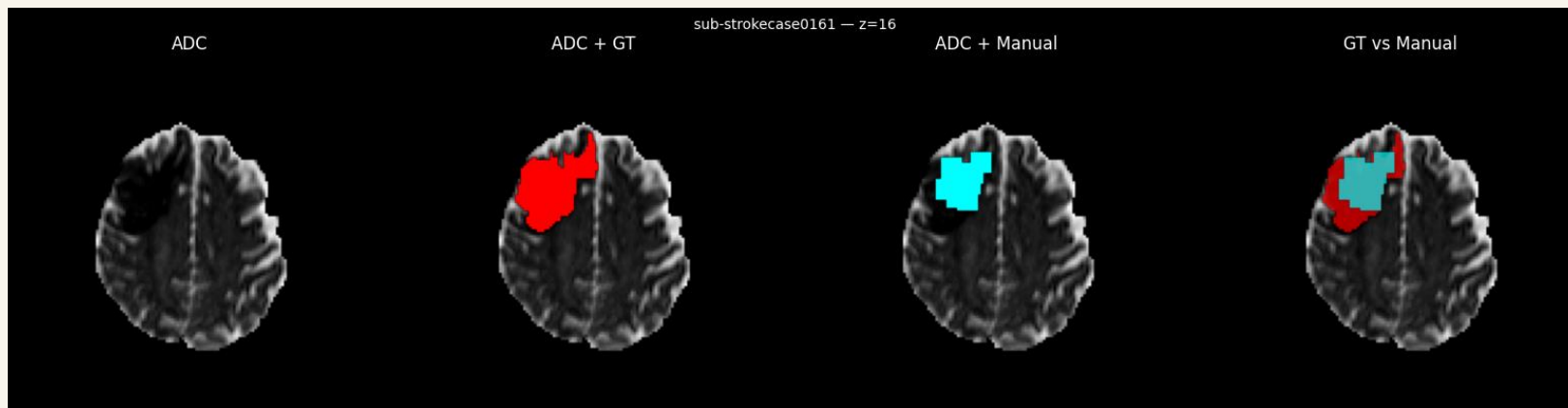


ISLES-2022 dataset composition (n=250)



Manual Segmentation

- **Tool:** ITK-SNAP, paintbrush tool, slice-by-slice on axial view
- **Data:** 10 cases from ISLES-2022, same subjects used for validation
- **Note:** No prior clinical experience therefore ground truth used as visual reference during annotation



Methods

Thresholding

- Simplest baseline and no training needed
- Labels voxels with abnormally low ADC as lesion
- Fails because subacute lesions have similar ADC to normal tissue

K-Means

- Unsupervised clustering which groups brain by ADC intensity
- Assumes lesion = darkest cluster
- No spatial awareness → segments nearly whole brain

2D U-Net

- Deep learning model which learns lesion patterns from 240 cases
- Encoder compresses, decoder reconstructs lesion mask
- Dice loss handles tiny lesion vs large background imbalance
- 50 epochs, Adam optimizer

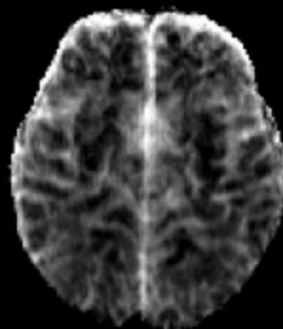
Results

SUMMARY (MEAN ± STD)

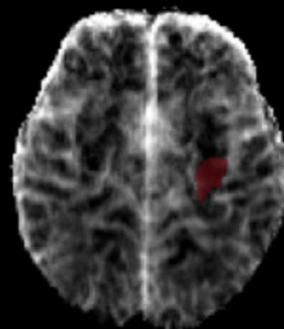
Method	Dice		HD95 (mm)	
	mean	std	mean	std
K-Means	0.038	0.049	41.47	18.05
Thresholding	0.064	0.087	40.52	18.06
U-Net	0.375	0.265	22.21	12.49

PER-SUBJECT DICE

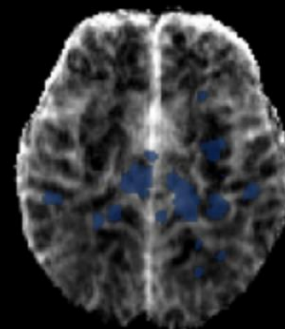
Subject	K-Means	Thresholding	U-Net
0137	0.005	0.010	0.005
0161	0.168	0.299	0.772
0195	0.032	0.055	0.452
0208	0.064	0.102	0.620
0216	0.002	0.002	0.040
0218	0.026	0.041	0.313
0219	0.023	0.036	0.377
0226	0.016	0.025	0.607
0228	0.018	0.031	0.078
0238	0.022	0.035	0.492



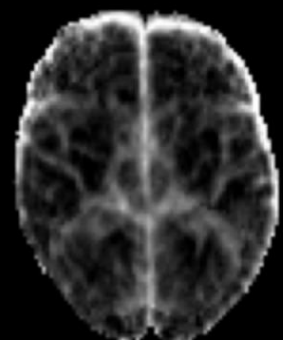
ADC MIP



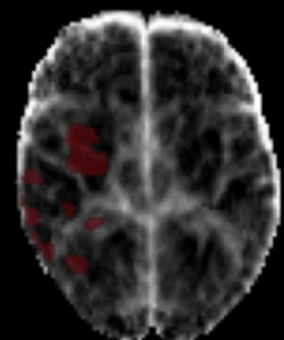
GT MIP
6.6 mL



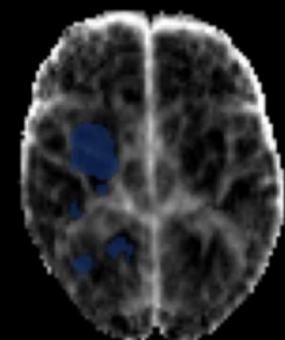
U-Net MIP
7.8 mL



ADC MIP



GT MIP
10.9 mL



U-Net MIP
2.2 mL

Discussion & Conclusion

- Thresholding & K-Means fail because subacute ADC overlaps normal tissue
- U-Net outperforms both classical methods by large margin
- U-Net struggles on small lesions → near-zero Dice
- High variance (std=0.265) → inconsistent across subjects
- Train loss drops & val loss flat → overfitting
- Intensity based methods not suitable for subacute stroke data
- U-Net viable but needs improvement

Future work could be:

- 3D U-Net → use spatial context across slices
- Data augmentation to reduce overfitting